Chapter 5. Instruction Set Architecture

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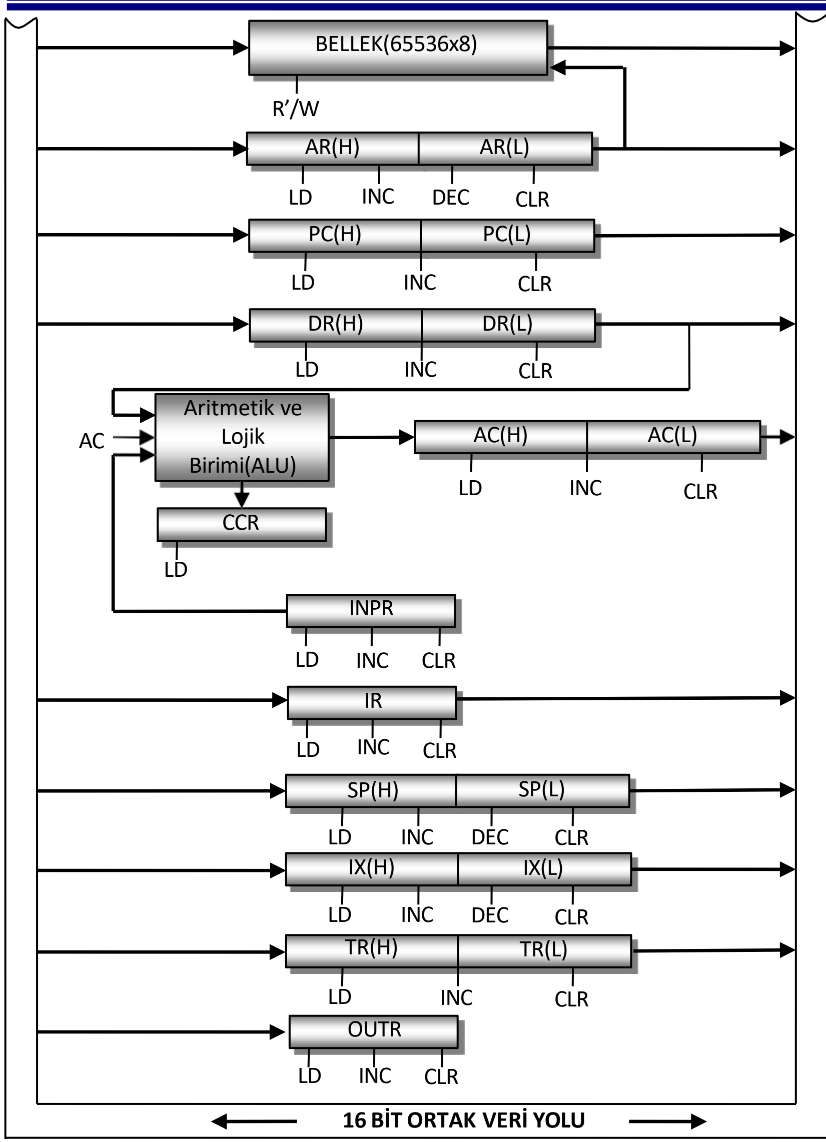
Components of Our Basic Computer System



Memory is considered as 64 KB. Each compartment of the memory holds 8 bits of data.

In addition, considering that some instructions may take up 1 byte in the instruction design, it was decided that each memory compartment should be 8-bit (Also, since the OPCODE part of the instructions is also 8-bit).

11 registers and memory are connected to the system bus. Which element will transfer information to the data bus in which situation is decided by the data arriving at the inputs of the 4×16 decoders.

Since the operations to be performed in the ALU will be 16 bits and the memory is 8 bits, the data bus is arranged in 2 parts; as the low and high significance part of the data path. For this reason, 2 decoders were used.

Allocation of Low and High Significance Bus

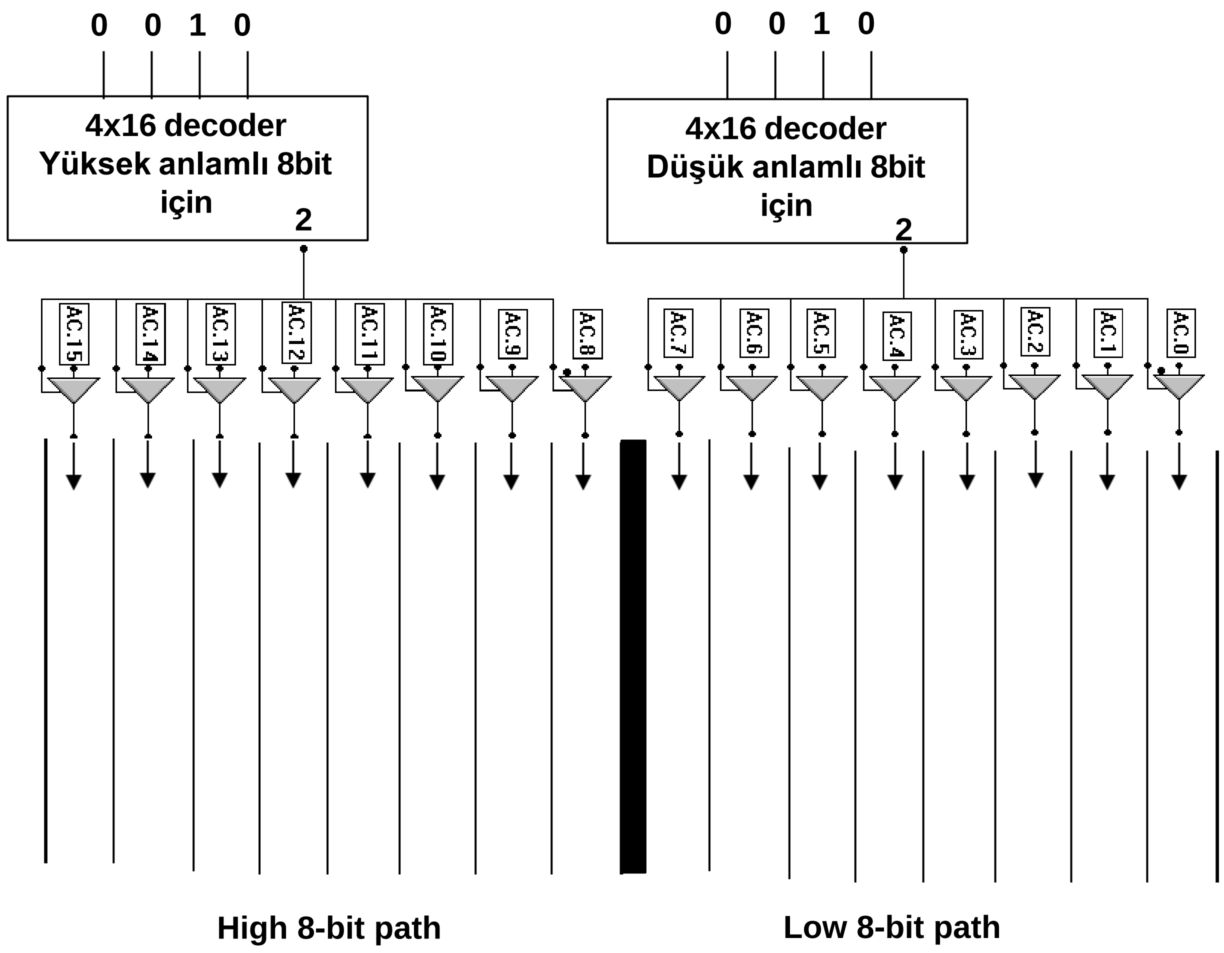
The binary values that must be applied to the inputs of the decoders in order to transfer the data of a register or memory to the bus are selected as in the table below.

Decoder Input	Componenet
0001	Stack Pointer (SP)
0010	Acumulator (AC)
0011	Program Counter (PC)
0100	Instruction Register (IR)
0101	Data Register (DR)
0110	Index Register (IX)
0111	Temporary Register (TR)
1000	Address Register (AR)
1001	Memory (MEM)
1010	For Index and Relative addressing
1100	Condition Code Register (CCR)

Registers except the status code register and command register are 16-bit. For 16-bit registers, a second decoder was used (High-significance bus).

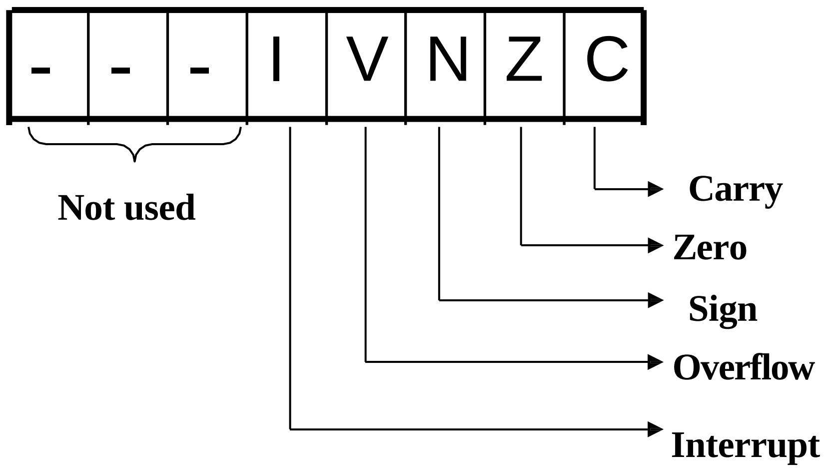
The input bits of the decoder that allocates the low-significance path/bus are the same as the input bits of the decoder that allocates the high-significance path.

Example Implementation: Transferring the Contents of the ACC to the Bus



Condition Code Register (CCR)

It works with ALU. It is also referred to as a flag register. The flag register, like all microprocessors, contains some information about the result of an operation, and this information has an effect on the instructions to be executed later. They are often used especially in executing decision-based commands.



Instruction Design

A program is a sequence of instructions that defines the operations the CPU must perform. It also shows the order in which the operations will be performed. A computer instruction is a binary code that defines a series of microprocessing steps.

Instructions reside in memory along with data. The computer reads each command from memory and writes it to the command register. The control unit decodes this code and executes the required micro operations respectively.

The command code consists of a group of bits. These tell the computer to perform a certain action. A command consists of several parts. The most important part of the command is the action part. The bits that make up the operation part define some operations such as addition, subtraction, multiplication, shift and complement. The number of operation bits in an instruction depends on the number of operations to be used in the computer. With n bits, 2n (or less) operations can be defined.

Another important part of the command code is the part that determines the addressing mode. In our basic computer system, 5 bits of the instruction code are reserved for the opcode. Although 32 operations can be performed with a five-bit opcode, since some instructions do not need some addressing modes, unused cases are allocated to other instructions and approximately 60 different operations can be defined.

Instruction Design

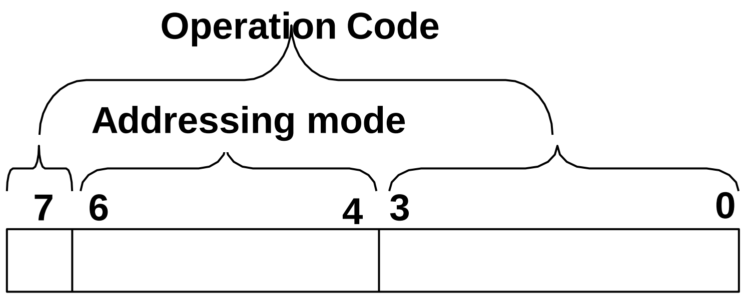
Since 6 types of addressing modes are used, 3 bits of the instruction code are allocated for the address space.

In our basic computer system, instructions with immediate, direct and indirect addressing modes are 3 bytes,

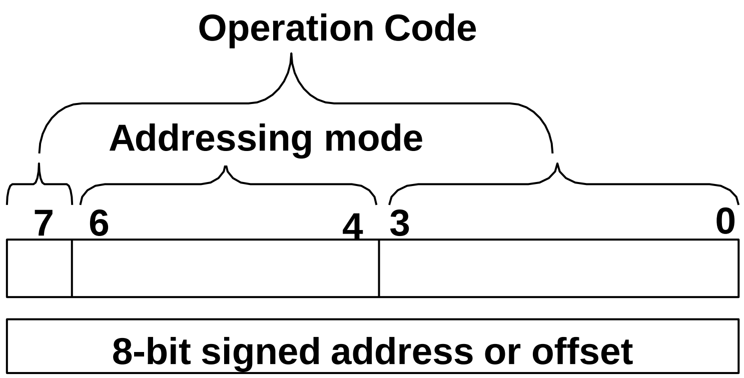
Instructions with index and relative addressing modes are 2 bytes,

Instructions with natural addressing mode take up 1 byte.

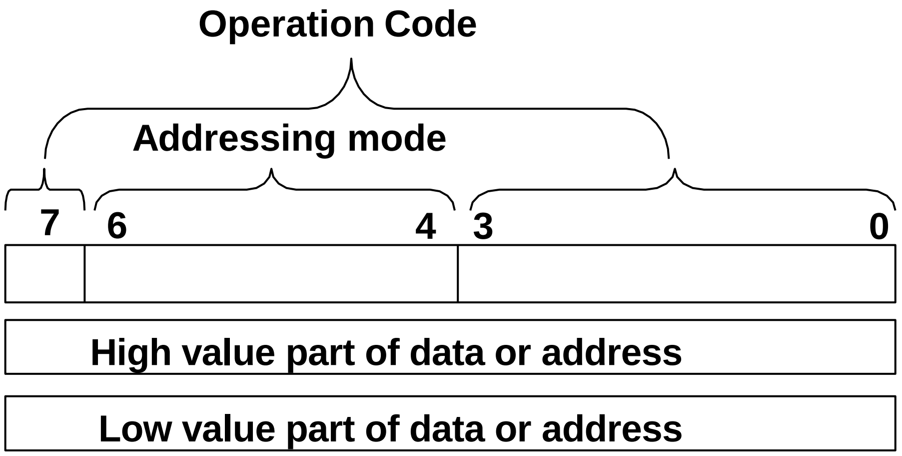
Instruction Format



Inherent addressing



Index and relative addressing



In immediate, direct and indirect addressing

Addressing Mode Selection Bits

3 bits are required to select the addressing mode.

Selection Bits	Addressing Modes
000	Inherent Mod
001	Immediate Mode
010	Direct Mode
011	Indirect Mode
100	Index Mode
101	Relative Mode

To ensure that the opcode changes sequentially when the addressing mode changes, 1 bit of the operation type is placed at the beginning of the opcode.

Example: In our basic computer system, the operation type of the addition (ADD) command is selected as 00000_2.

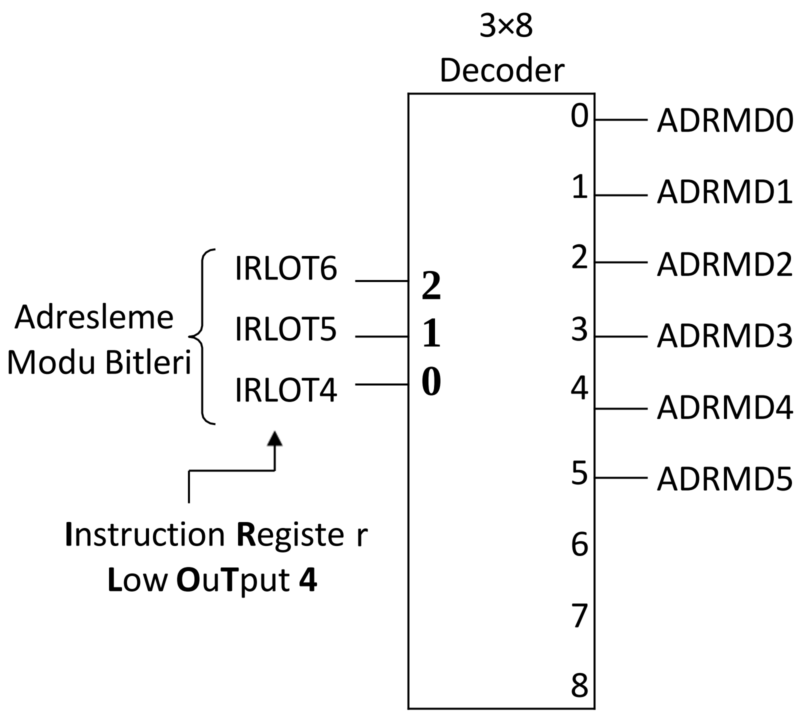
Opcode part for immediate addressing; | 0 001 | 0000 | Opcode equivalent 10h

Opcode part for direct addressing; | 0 010 | 0000 | Opcode equivalent 20h

Resolving Addressing Mode Information

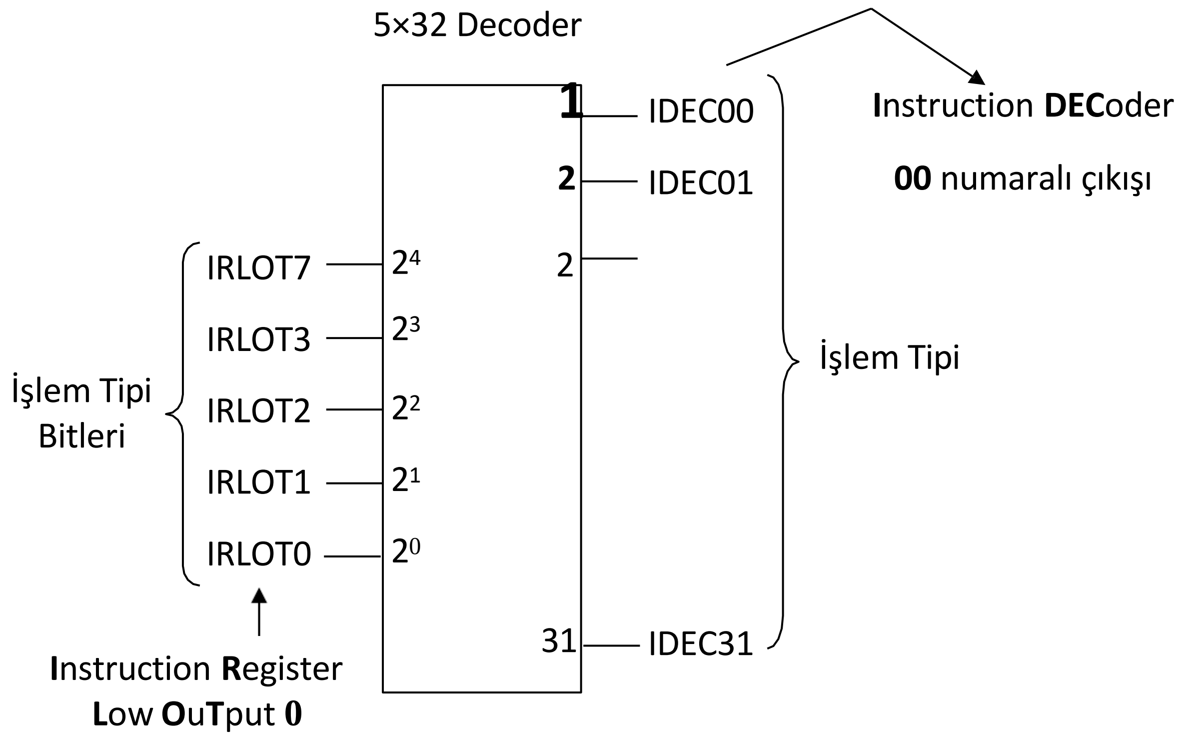
While the commands in memory are executed, the opcode parts are transferred to the instruction register. Bits 4, 5 and 6 of the opcode serve to determine the addressing mode. These three bits were passed through a 3×8 decoder and 8 outputs were obtained. 6 of them are used in our basic computer system. Addressing modes are named with the following abbreviations;

Control Name	Addressing Mode
ADRMD0	Inherite Mode
ADRMD1	Immediate Mode
ADRMD2	Direct Mode
ADRMD3	Indirect Mode
ADRMD4	Index Mode
ADRMD5	Relative Mod



Analyzing the Operation Type Information of the Instruction

Bits 0, 1, 2, 3 and 7 of the opcode in the instruction register are used to determine the operation type. These five bits were passed through a 5×32 decoder and 32 outputs were obtained.



Parsing the Instruction

The command is decoded according to the signals coming from the hardware mechanisms that solve the addressing mode and operation type.

For example, let's assume that the 5 bits indicating the operation type in the instruction register are 00101_2. In this case, output number 5 of the decoder that decodes the operation type, that is, the IDEC05 signal, will be active.

This signal alone will not be sufficient to determine the type of command. Because there may be other commands that produce the same signal. When creating the instruction set of our basic computer system, the same operation type was chosen for both the division instruction (DIV) and the two's complement retrieval instruction (NEG). Because the division instruction does not use the inherite addressing mode (since it always receives operand information). In other words, the NEG instruction is defined with the combination of operation type 5 and addressing mode 0.

Parsing the Instruction

Instruction	**Opcode**				
	Inherit	Immediate	Direct	Indirect	Index
DIV	-	15h	25h	35h	45h
NEG	05h	-	-	-	-

IDEC05.ADRMD0 condition will activate the NEG command,

IDEC05.ADRMD1
IDEC05.ADRMD2
IDEC05.ADRMD3
IDEC05.ADRMD4

Any of the conditions will activate the DIV command.

These activation signals will be used to activate the hardware mechanisms related to the execution of commands.